

Michael P. O'Donnell, Ph.D.

Assistant Professor
Molecular, Cellular and Developmental Biology

Education

University of Pennsylvania
Drexel University

Philadelphia, PA
Philadelphia, PA

Cell & Molec. Bio
Bio. Engineering

Ph.D., 2012
B.S., 2002

Academic appointments

Assistant Professor

Yale University, Dept. of Molecular, Cellular and Developmental Biology

2021-present
New Haven, CT

Postdoctoral Fellow

Brandeis University, Dept. of Biology
Advisor: Piali Sengupta

2013 - 2020
Waltham, MA

Graduate student

University of Pennsylvania School of Medicine, program in Cell and Molecular Biology
Advisor: Greg Bashaw

2006-2012
Philadelphia, PA

Research technician

University of Pennsylvania School of Medicine, Dept. of Hematology-Oncology
Advisor: Peter S. Klein

2004-2006
Philadelphia, PA

Research technician

University of Pennsylvania School of Vet. Medicine
Advisor: Jean-Pierre Saint-Jeannet

2002-2004
Philadelphia, PA

Publications (*corresponding author)

Complete List of Published Work in MyBibliography:

<https://www.ncbi.nlm.nih.gov/myncbi/michael.o'donnell.2/bibliography/public/>

Preprints and manuscripts in review

1. Porhathai Malaiwong, Allen F. Schroeder, Tia Brown, Chester Wrobel, Nawaphat Malaiwong, Madhumanti Dasgupta, Frank C. Schroeder, and **Michael P. O'Donnell***, Nuclear receptor-neurotransmitter coupling links behavior to metabolic state, *bioRxiv* doi: <https://doi.org/10.1101/2025.10.20.683577>, in revision: *Nature* (2025).
2. Christina Campagna, Abigail Descoteaux, Abigail Poole, Eric Peet, Nawaphat Malaiwong, **Michael P. O'Donnell**, and Inna Nechipurenko, Cell-specific roles for the conserved galpha chaperone ric-8 in cilia biology, *bioRxiv* doi: <https://doi.org/10.64898/2026.02.07.704597>, in revision: *MBOC* (2026).
3. Rong Xuan Zang, Nawaphat Malaiwong, Ling Wang, Jamie D. Maziarz, Kejue Jia, Bernhard Drotleff, Frank Stein, Mah Noor, C. Jackson Roberts, Mandy Rettel, Jennifer J. Schwarz, Aissam Ikmi, Shigeki Watanabe, Robert Prevedel, **Michael P. O'Donnell**, and Jacob M. Musser, An ancient monoaminergic signaling system coordinates contractility in a nerveless sponge, *bioRxiv* .

Peer reviewed primary research papers

4. Nawaphat Malaiwong, Porhathai Malaiwong, Chloe Kim, and **Michael P. O'Donnell***, Flint 2.0: Robust and customizable single shot integration in *C. elegans*, G3 (in press) doi: <https://www.biorxiv.org/content/10.64898/2026.01.12.699147v1> (2026).
5. Alison Philbrook, **Michael P. O'Donnell**, Laura Grunenkovaitė, and Piali Sengupta, Cilia structure and intraflagellar transport differentially regulate sensory response dynamics within and between *C. elegans* chemosensory neurons, *PLOS Biology* **22**, e3002892 (2024).
6. Munzareen Khan, Anna H. Hartmann, **Michael P. O'Donnell**, Madeline Piccione, Anjali Pandey, Pin-Hao Chao, Noelle D. Dwyer, Cornelia I. Bargmann, and Piali Sengupta, Context-dependent reversal of odorant preference is driven by inversion of the response in a single sensory neuron type, *PLOS Biology* **20**, e3001677 (2022).

7. Travis Kyani-Rogers, Alison Philbrook, Ian G. McLachlan, Steven W. Flavell, **Michael P. O'Donnell**, and Piali Sengupta, Developmental history modulates adult olfactory behavioral preferences via regulation of chemoreceptor expression in *Caenorhabditis elegans*, *Genetics* **222**, iyac143 (2022).
8. Chester J. J. Wrobel, Jingfang Yu, Pedro R. Rodrigues, Andreas H. Ludewig, Brian J. Curtis, Sarah M. Cohen, Bennett W. Fox, **Michael P. O'Donnell**, Paul W. Sternberg, and Frank C. Schroeder, Combinatorial Assembly of Modular Glucosides via Carboxylesterases Regulates *C. elegans* Starvation Survival, *Journal of the American Chemical Society* **143**, 14676–14683 (2021).
9. Jodie A. Schiffer, Stephanie V. Stumbur, Maedeh Seyedolmohadesin, Yuyan Xu, William T. Serkin, Natalie G. McGowan, Oluwatosin Banjo, Mahdi Torkashvand, Albert Lin, Ciara N. Hosea, Adrien Assié, Buck S. Samuel, **Michael P. O'Donnell**, Vivek Venkatachalam, and Javier Apfeld, Modulation of sensory perception by hydrogen peroxide enables *Caenorhabditis elegans* to find a niche that provides both food and protection from hydrogen peroxide, *PLOS Pathogens* **17**, e1010112 (2021).

Publications prior to arrival at Yale

10. **Michael P. O'Donnell***, Bennett W. Fox, Pin-Hao Chao, Frank C. Schroeder, and Piali Sengupta, A neurotransmitter produced by gut bacteria modulates host sensory behaviour, *Nature* **583**, 415–420 (2020), Pubmed Central PMCID: PMC7853625.
11. **Michael P. O'Donnell***, Pin-Hao Chao, Jan E. Kammenga, and Piali Sengupta, Rictor/TORC2 mediates gut-to-brain signaling in the regulation of phenotypic plasticity in *C. elegans*, *PLoS Genetics* **14**, e1007213 (2018), Pubmed Central PMCID: PMC5819832.
12. Scott J. Neal, Asuka Takeishi, **Michael P. O'Donnell**, JiSoo Park, Myeongjin Hong, Rebecca A. Butcher, Kyuhyung Kim, and Piali Sengupta, Feeding state-dependent regulation of developmental plasticity via CaMKI and neuroendocrine signaling, *eLife* **4** (2015), Pubmed Central PMCID: PMC4558564.
13. **Michael P. O'Donnell** and Greg J. Bashaw, Distinct functional domains of the Abelson tyrosine kinase control axon guidance responses to Netrin and Slit to regulate the assembly of neural circuits, *Development* **140**, 2724–2733 (2013), Pubmed Central PMCID: PMC3678342.
14. **Michael P. O'Donnell** and Greg J. Bashaw, Src inhibits midline axon crossing independent of Frazzled/Deleted in Colorectal Carcinoma (DCC) receptor tyrosine phosphorylation, *The Journal of Neuroscience* **33**, 305–314 (2013), Pubmed Central PMCID: PMC3739878.
15. David S. Garbe, **Michael P. O'Donnell**, and Greg J. Bashaw, Cytoplasmic domain requirements for Frazzled-mediated attractive axon turning at the *Drosophila* midline, *Development* **134**, 4325–4334 (2007), Pubmed ID PMID: 18003737.
16. **Michael P. O'Donnell**, Chang-Soo Hong, Xiao Huang, Raymond J. Delnicki, and Jean-Pierre Saint-Jeannet, Functional analysis of Sox8 during neural crest development in *Xenopus*, *Development* **133**, 3817–3826 (2006), Pubmed ID PMID: 16943273.
17. Jinling Wu, **Michael O'Donnell**, Aaron D. Gitler, and Peter S. Klein, Kermit 2/XGIPC, an IGF1 receptor interacting protein, is required for IGF signaling in *Xenopus* eye development, *Development* **133**, 3651–3660 (2006), Pubmed ID PMID: 16914488.
18. Xiao Huang, Chang-Soo Hong, **Michael O'Donnell**, and Jean-Pierre Saint-Jeannet, The doublesex-related gene, XDmrt4, is required for neurogenesis in the olfactory system, *Proceedings of the National Academy of Sciences* **102**, 11349–11354 (2005), Pubmed Central PMCID: PMC1183594.

Publications: reviews and commentaries

19. **Michael P. O'Donnell**, Munzareen Khan, and Piali Sengupta, Thermosensation: Human parasitic nematodes use heat to hunt hosts, *Current Biology* **28**, R795–R798 (2018), Pubmed ID PMID: 30040942.
20. **Michael P. O'Donnell**, Rebecca K. Chance, and Greg J. Bashaw, Axon growth and guidance: Receptor regulation and signal transduction, *Annual Review of Neuroscience* **32**, 383–412 (2009).

Fellowship and Grant Support

• Current

NIH DP2GM154014-01 New Innovator award (NIGMS) Role: PI
Molecular determinants of host-feeding manipulation and microbial colonization
Direct Costs: \$1,500,000 Indirect costs: \$843,751

09/01/2023 - 08/31/2028

• Pending

NIH 1R01DC023551-01 Role: PI
A novel class of glucosylated neuromodulators which encode dietary history
to regulate chemosensory behaviors
Direct Costs: \$1,250,000 Indirect costs: \$754,355

proposed: 01/01/2026 - 12/31/2030

- **Completed**

1F32DC013711-01

Genetic and neural basis of pheromone sensory integration in nematodes
Direct Costs: \$98,428

01/01/2014 - 12/31/2015

Teaching

Yale College courses

Semester(s)

Genetics and Development (BIOL 1030)

S2026, S2024, S2023, F2021

Foundation lecture course for all Bioscience majors

Semester	Enrollment	Lectures
Spring 2026	165	12 x 75 min (Course director)
Spring 2024	188	12 x 75 min (Course director)
Spring 2023	190	12 x 75 min (Course director)
Fall 2021	122	2 x 75 min

Sensory Neuroscience Through Illusions (MCDB 3290/NSCI 3290)

F2025, F2024, F2023, F2022

Lecture course

Semester	Enrollment	Lectures
Fall 2025	19	5 x 75 min
Fall 2024	47	4 x 75 min
Fall 2023	59	4 x 75 min
Fall 2022	17	4 x 75 min

Brain Development and Plasticity (MCDB 4400)

S2026

Seminar course

Semester	Enrollment	Lectures
Spring 2026	21	1 x 110 min

Independent research (MCDB 4740,4750,4850,4860,4950,4960)

S2022, F2021, S2021

Supervision of undergraduate research course

Semester	Enrollment
Spring 2022	91(20, 3, 4, 53, 2, 9)
Fall 2021	88(16, 3, 55, 5, 9, 0)
Spring 2021	53(6, 1, 5, 32, 0, 14)

Yale University Graduate teaching

Semesters(s)

Biology of Bacterial Pathogens II (MBIO 6850)

S2026, S2025, S2024, S2023, S2021

Graduate lecture and seminar course

Semester	Enrollment	Lectures
Spring 2026	13	2 x 90 min
Spring 2025	3	2 x 90 min
Spring 2024	7	2 x 90 min
Spring 2023	6	2 x 90 min
Spring 2021	6	2 x 90 min

Interdepartmental Neuroscience Program Coding Bootcamp

F2025, F2024, F2023, F2022

1 day behavioral analysis module for incoming INP students

Semester	Enrollment	Lectures
Fall 2025	17	1 day module
Fall 2024	23	1 day module
Fall 2023	21	1 day module
Fall 2022	22	1 day module

Responsible Conduct of Research (MCDB 5040)

S2026

Required refresher course in research ethics for GSAS students

<i>Semester</i>	<i>Enrollment</i>	<i>Lectures</i>
Spring 2026	40	90 min

Research in Progress seminars (MCDB 9020)

F2024

Yearly progress update for all MCDB PhD students

<i>Semester</i>	<i>Enrollment</i>	<i>Lectures</i>
Fall 2024	50	2 half day sessions

Training program in Genetics ethics workshop

S2025

Case study reviews and discussion for TPG PhD trainees

<i>Semester</i>	<i>Enrollment</i>	<i>Lectures</i>
Spring 2025	17	120 min

Service

MCDB departmental service

• Chair (interim), MCDB Community and Professional Climate committee	2022, Spring 2026
• Committee member, MCDB open faculty search	2022, 2025
• Committee member, MCDB Graduate Affairs Committee	2024, 2025, 2026
• Organizer, MCDB Graduate Research in Progress symposia	2024
• Admissions committee, MCGD/BBS graduate program	2021, 2022, 2023, 2024
• Organizer, MCDB annual retreat	2022, 2023
• Member, MCDB Community and Professional Climate committee	2021 -
• Organizer, MCDB Faculty research in progress seminars	2021-2022

Yale College and University contributions

• At large member, Learning Environment Subcommittee (LESC), YSM Education Policy and Curriculum Committee	2023-
• Organizer, C. elegans Bi-weekly Seminar Series (O'Donnell, Bahmanyar, Reinke, Koelle, Hammarlund, Yorgev, Colón-Ramos, Davis and Bennett labs)	2021-
• Member, Wu Tsai Institute	2022-
• Reviewer, Yale College Dean's Research Fellowship	2022, 2023, 2024, 2025, 2026
• Participant, MCGD/BBS interview and recruitment weekends	2021-
• Selection committee, BBS Tatum graduate fellowships	2024
• Selection committee, Kavli awards for undergraduate neuroscience research	2022, 2023, 2024
• Advisor, Yale MCDB and Neuroscience undergraduate majors	2021-
• Selection committee, Boell award for undergraduate MCDB research	2022, 2023
• Interviewer, Endocrinology Medical Fellow program, Internal Medicine	2021

Doctoral candidacy examination committees

• Giramnah Peña-Alcántara	Huckins lab, Psychiatry/INP
• Sarah Morgan	Musser lab, MCDB
• Roy Zang	Musser lab, MCDB
• Taylor Peterson	Carlson lab, MCDB
• Heng Wu	Clark lab, MCDB
• Sisi Zheng	Crews lab, MCDB

- Evyn Dickinson
- Emily Siff
- Efrat Abramson

Jeanne lab, Neuroscience/INP
 Carlson lab, MCDB/INP
 Sansing lab, Neurology/INP

Thesis advisory committees

• Sarah Morgan (committee chair)	Musser lab, MCDB
• Roy Zang (committee chair)	Musser lab, MCDB
• Taylor Peterson	Carlson lab, MCDB
• Heng Wu	Clark lab, MCDB
• Emily Siff	Carlson lab, MCDB/INP
• Emerson Santiago	Koelle lab, MBB/MCDB
• Max Bales, BS/MS	van Wolfswinkel lab, MCDB
• Alicia Stein, PhD (2024)	Crews lab, MCDB
• Sisi Zheng, PhD (2024 - committee chair)	Crews lab, MCDB

Professional/society service

• Organizer, 3rd Annual C. elegans microbiome workshop, International C. elegans conference	2021
• Ad hoc reviewer for multiple journals	
• Member, Society for Neuroscience	2012-
• Member, Society for Developmental Biology	2006-2013
• Member, Genetics Society of America	2007-

Career counseling

Postdoctoral fellows

Name	Lab	Institution	Program	Years
Lisa Baik	Carlson (MCDB)	Yale University	MCDB mentorship	2023 - 2025
Shoken Lee	Bahmanyar (MCDB)	Yale University	MCDB mentorship	2023 - 2025
Geoffrey Thomson	Jacob (MCDB)	Yale University	MCDB mentorship	2024 -

Graduate students

Rachael Assaolu	Srinivasan Lab	WPI	C. elegans mentoring	2025-
Adam Friedberg	Hart Lab	Brown University	C. elegans mentoring	2024

Mentoring

O'Donnell Lab current trainees

Graduate students

Name	Role	Dept./Program	Years
Patty Benitez-Lomi	Doctoral student	MCDB (BBS-MCGD)	2024 -

Postdoctoral fellows

Name	Dept./Program	Years
Sebastian Gonzalez La Corte	MCDB	2025 -
Nawaphat Malaiwong	MCDB	2024 -

Undergraduate students

Name	Role	Institution	Program	Years
Sarah Mahmoud ('26)	Senior thesis	Yale College	MCDB	2024 -
Caroline Utermann ('26)	Senior thesis	Yale College	Neuroscience	2024 -
Patricia Joseph ('26)	Senior thesis (STARS II)	Yale College	MCDB	2024 -

Postgraduates

Name	Years
Porhathai Malaiwong	2024 -

O'Donnell Lab former trainees

Undergraduate students

Name	Institution	Program	Years	Current role
Adriana Galindo Oelsner	U of Toronto	Independent	2024	Yale BBS PhD prog.
Mackenzie Altuzar ('27)	Yale College	MCDB (STARS)	2024	Yale College
Christian Guirgis ('26)	Yale College	MCDB (Summer)	2023	Yale College
Dina Garmroudi ('24)	Yale College	MCDB (STARSII)	2022 - 2024	Johns Hopkins Med School
Ahlem Dorgham	Yale NUS	Dean's Fellowship	2022	Research Assistant, YALE NUS
Julia Balch ('23)	Yale College	MCDB (STARSII)	2021 - 2023	Princeton MolBio PhD
Andrew Jackson III ('22)	Yale College	MCDB	2021 - 2022	NYU Grossman Med

Postgraduates

Name	Years	Current role
Tia Brown	2021 - 2024	applying for PhD programs

Postdocs

Name	Years	Current role
Madhumanti Dasgupta	2021 - 2025	applying for academic positions

Graduate students

Name	Years	Current role
Jason Bishai	2022 - 2024	Medical leave/withdrawn

Rotation students

Name	Role	Institution	Program	Years
Chloe Kim	Rotation student	Yale University	MCGD (BBS)	2025
Lex Herrerias	Rotation student	Yale University	INP (BBS)	2023
Giramnah Peña-Alcántara	Rotation student	Yale University	INP (BBS)	2023
Nora Nelson	Rotation student	Yale University	Microbiology (BBS)	2022
Taylor Peterson	Rotation student	Yale University	MCGD (BBS)	2022
Heng Wu	Rotation student	Yale University	MCGD (BBS)	2022
JoAnne Villagrana	Rotation student	Yale University	MCGD (BBS)	2022
Matthew Steinsaltz	Rotation student	Yale University	BQBS (BBS)	2021

Awards

Awards for Mike O'Donnell, Ph.D.

• NIH New Innovator Award	2023-2028
• RCSA Scialog Fellow	2022-2023
• Rosbash-Abovich Molecular and Cellular Biology Excellence Award	2020
• Susan Lindquist Fund award	2019
• Ruth L. Kirschstein (NRSA) <i>Genetic and Neural Basis of Pheromone Sensory Integration in Nematodes</i> NIH F32 DC013711	2014-2015
• Training program in Quantitative Neuroscience NIH T32-NS-007292	2013
• Training program in Developmental Biology NIH T32-HD007516	2009-2012
• Training program in Cell and Molecular Biology NIH T32-GM07229	2007-2009
• AJ Drexel Scholarship	1997-2002

Awards for O'Donnell lab trainees

• Caroline Utermann, Gates Cambridge Fellowship award	2026
• Patricia Joseph, ABRCMS award for top poster presentation	2025
• Julia Balch, Boell thesis prize for top senior thesis	2023

Invited Presentations

Invited seminars presented by Mike O'Donnell, Ph.D.

<i>Going with your gut: how C. elegans uses chemistry to sense the microbes within</i> University of Tennessee	2026
<i>Going with your gut: how C. elegans uses chemistry to sense the microbes within</i> Max Planck Institute for Biology Tübingen and the Friedrich Miescher Laboratory Student Invited Distinguished Speaker Seminar Series	2025
Keynote lecture <i>Going with your gut: how C. elegans uses chemistry to sense the microbes within</i> University of Pennsylvania DSRB retreat	2024
<i>Going with your gut: how C. elegans uses chemistry to sense the microbes within</i> University of Texas, Southwestern, Excellence in Immunology Lecture Series	2024
International Behavioral and Neural Genetics Society (IBANGS) conference: Genes, Brain and Behavior Symposium: Genes, Microbes, and Behavior: Insights from Non-Mammalian Models - canceled	2023
<i>Going with your gut: how C. elegans uses chemistry to sense the microbes within</i> Rockefeller University, Kavli Neural Systems Institute symposium Microbiota and the Brain: impact on nervous system development, function, and evolution	2023
<i>Modulation of host behavior by gut microbial metabolites</i> University of Massachusetts Chan Medical School, Neuroscience Program seminar series	2023
<i>Modulation of host behavior by gut microbial metabolites</i> Indiana University, Microbiology seminar series	2023
<i>Gut bacteria metabolites and host metabolism converge to regulate sensory behaviors</i> Syracuse University Biology Dept seminar series	2022
<i>Modulation of host behavior by gut microbial metabolites</i> University of Kent, Canterbury. School of Biosciences seminar series	2022
<i>Bacterially produced neuromodulators alter C. elegans olfactory behaviors</i> Yale University Microbiology Retreat	2021

Rosbash-Abovich award lecture

<i>A neurotransmitter produced by gut bacteria modulates host sensory behaviour</i> Brandeis University	2020
<i>Modulation of C. elegans behavior by gut microbes</i> World Wide Neuro seminar series, Brain-Body Interactions	2020
<i>Microbial metabolites regulate host behaviors</i> MCDB retreat, Yale University	2020
<i>Modulation of aversive chemical responses by tyramine-producing bacteria</i> Volen Center for complex systems retreat, Brandeis University	2019
<i>TORC2 signaling and gut colonization regulates neuronal state</i> Sommer laboratory, Max Planck Institute, Tübingen, Germany	2017
<i>Natural variation in a TOR-complex 2 component underlies a temperature-dependent polyphenic trait</i> International Worm Meeting Quantitative Genetics Workshop, UCLA	2015

Platform presentations by O'Donnell lab trainees

Selected talk, Nawaphat Malaiwong, Ph.D., Postdoctoral Fellow <i>International C. elegans Conference, UC Davis</i> <i>Emerging Genetic Tools for C. elegans</i>	2025
Selected talk, Julia Balch ('23), Yale undergraduate student <i>Gordon Research Conference on Animal-Microbe Symbiosis, Lucca, Italy</i>	2023
Selected talk, Madhumanti Dasgupta, Ph.D., Postdoctoral Fellow <i>International C. elegans Conference, Glasgow, Scotland, UK</i>	2023

Presentations by Mike O'Donnell, Ph.D. prior to arrival at Yale

O'Donnell, MP , Fox, BW, Chao, PH, Schroeder, FC and Sengupta, P <i>Modulation of aversive chemical responses by tyramine-producing bacteria</i> <i>C. elegans Neuro meeting, UW Madison</i>	2019
O'Donnell, MP , Chao, PH, Kammenga, JE and Sengupta, P <i>Modulation of phenotypic plasticity by gut-brain signaling</i> Gordon Research Conference on Modulation of Neural Circuits and Behavior, Sunday River, ME	2018
O'Donnell, MP , Chao, PH, Kammenga, JE and Sengupta, P <i>Intestinal mTORC2 and gut colonization regulates internal state</i> Boston Area Worm Meeting (BAWM), MIT, Cambridge, MA	2017
O'Donnell, MP , Chao, PH and Sengupta, P <i>Intestinal colonization by bacteria alters chemosensory responses to alcohols</i> International Worm Meeting Pathogen parallel session, UCLA	2017